QUALITY INFORMATION DOCUMENT

For North Pacific Gyre Oscillation OMI_VAR_EXTREME_CLIMVAR_PACIFIC_npgo_sla _eof_mode_projection

Issue: 4.0

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RECORD TABLE

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The fourth column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	2019/01/21	All	Creation	B. Tranchant	MI Pujol
1.1	2020/03/25	Ref	Reference update	J.-F. Legeais	
1.2	2020/05/06	Annex A	Addition of the Ocean State Report #3, section 2.6	J.-F. Legeais	
2.0	2022/04/04	All	Update template; typos	C. Pegliasco	MI Pujol
2.1	2023/04/05		Update upstream products list	MI Pujol	MI Pujol
3.0	2023/06/15		OMI nomenclature change Use « myint » dataset	MI Pujol	MI Pujol
4.0	2025/05/28		Reprocessing DT2024	MI Pujol	K Grassi

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1) INTRODUCTION

a) Product Name

OMI_VAR_EXTREME_CLIMVAR_PACIFIC_npgo_sla_eof_mode_projection.

b) Indicator Evaluation Method

The North Pacific Gyre Oscillation (NPGO) is a climate pattern introduced by Di Lorenzo et al. (2008). The NPGO is defined as the second dominant mode of variability of Sea Surface Height (SSH) anomaly in Northeast Pacific (25°–62°N, 180°–250°E, see Figure 2 of Di Lorenzo et al. (2008). From the principal component analysis, the EOF's amplitude pattern or correlation map has been inferred and projected on the SSH anomalies from an historical eddy-permitting model hindcast of the Northeast Pacific over the period 1950-2004.
From 2004 onward, the CMEMS NPGO index OMI is derived from the last version (DT-2024, see https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-SL-QUID-008-032-068.pdf) of the reprocessed altimeter sea level delayed-time record (SEALEVEL_GLO_PHY_L4_MY_008_047) and from the near real time altimeter sea level (SEALEVEL_GLO_PHY_L4_NRT _008_046).
Sea level change time series have a strong seasonal signal and long-term trend. The altimeter data record is used from year 1993 to fit a model on annual, semi-annual sea level variability, and linear trend changes. This model is then used to remove these signals on each grid points of the altimeter map series. The temporal variations corresponding to the NPGO EOF mode are deduced from the series of sea level grid obtained. The NPGO index obtained is centered and normalized to fit with the Di Lorenzo index (Di Lorenzo et al., 2008), over the [1993, 2003] period. Finally, NPGO index deduced from altimetry in completed with historical index from Di Lorenzo et al (2008) from 2004 downward.

A detailed description of the North Pacific Gyre Oscillation index has been provided by Tranchant et al. (2019) in section 2.6 of the CMEMS Ocean State Report #3 and this section is included in **Erreur ! Source du renvoi introuvable.** of this document.

c) Copernicus Marine products used as input for the computation

Product name (and Type)	Documentation
SEALEVEL_GLO_PHY_L4_MY_008_047 SEALEVEL_GLO_PHY_L4_NRT_008_046	QUID : https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-SL-QUID-008-032-068.pdf PUM : https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-SL-PUM-008-046-047-060-068.pdf

2) SCIENTIFIC VALIDATION

The North Pacific Gyre Oscillation (NPGO) term is used because its fluctuations reflect changes in the intensity of the central and eastern branches of the North Pacific gyres circulations Chhak et al.2009, « Forcing of Low-Frequency Ocean Variability in the Northeast Pacific »..
Di Lorenzo et al. (2008) suggested that the North Pacific Gyre Oscillation is the oceanic expression of the atmospheric variability of the North Pacific Oscillation Walker et Bliss, *World Weather V.*, which has an expression in both the 2nd EOFs of SSH and Sea Surface Temperature (SST) anomalies Ceballos et al., 2009 « North Pacific Gyre Oscillation Synchronizes Climate Fluctuations in the Eastern and Western Boundary Systems ».. This finding is further supported by the recent work of Yi et al. (2018) showing consistent pattern features between the atmospheric North Pacific Oscillation and the oceanic North Pacific Gyre Oscillation.
This index measures change in the North Pacific gyres circulation and explains key physical-biological ocean variables including temperature, salinity, sea level, nutrients, chlorophyll-a. One example is the coherent variation of modeled and observed SSS (Sea Surface Salinity) in the North East Pacific Di Lorenzo et al., 2009 « Nutrient and Salinity Decadal Variations in the Central and Eastern North Pacific »; Di Lorenzo et al., « North Pacific Gyre Oscillation Links Ocean Climate and Ecosystem Change ».. Like the Pacific Decadal Oscillation (PDO), the North Pacific Gyre Oscillation is a basin-scale feature and captures prominent low-frequency changes in the Pacific physical and biological fields in the Kuroshio western boundary current system Ceballos et al., 2009 « North Pacific Gyre Oscillation Synchronizes Climate Fluctuations in the Eastern and Western Boundary Systems ». and the central tropical Pacific Di Lorenzo et al., 2010 « Central Pacific El Niño and Decadal Climate Change in the North Pacific Ocean ».. Thus, the North Pacific Gyre Oscillation index provides a strong indicator of fluctuations in the mechanisms driving planktonic ecosystem dynamics.
A positive North Pacific Gyre Oscillation phase is a dipole pattern with negative SSH anomaly north of 40°N and a positive SSH anomaly south of 40°N.
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In Nov 2025, a new version of the OMI was delivered. It corresponds to a reprocessing of the series using the up-to-date version of the upstream DUACS products (DT-2024 version). The change in the reprocessed version (previously DT-2021) induces some slight changes not impacting the general variability of the OMI (correlation coefficient of 0.9995 for the [1950-2024] period).

3) REFERENCES

Ceballos, Lina I., Emanuele Di Lorenzo, Carlos D. Hoyos, Niklas Schneider, et Bunmei Taguchi. « North Pacific Gyre Oscillation Synchronizes Climate Fluctuations in the Eastern and Western Boundary Systems ». *Journal of Climate* 22, n° 19 (1 octobre 2009): 5163-74. https://doi.org/10.1175/2009JCLI2848.1.

Chhak, Kettyah C., Emanuele Di Lorenzo, Niklas Schneider, et Patrick F. Cummins. « Forcing of Low-Frequency Ocean Variability in the Northeast Pacific ». *Journal of Climate* 22, n° 5 (1 mars 2009): 1255-76. https://doi.org/10.1175/2008JCLI2639.1.

Di Lorenzo, E., K. M. Cobb, J. C. Furtado, N. Schneider, B. T. Anderson, A. Bracco, M. A. Alexander, et D. J. Vimont. « Central Pacific El Niño and Decadal Climate

Change in the North Pacific Ocean ». *Nature Geoscience* 3, n° 11 (novembre 2010): 762-65. https://doi.org/10.1038/ngeo984.

Di Lorenzo, E., J. Fiechter, N. Schneider, A. Bracco, A. J. Miller, P. J. S. Franks, S. J. Bograd, et al. « Nutrient and Salinity Decadal Variations in the Central and Eastern North Pacific ». *Geophysical Research Letters* 36, n° 14 (2009). https://doi.org/10.1029/2009GL038261.

Di Lorenzo, E., N. Schneider, K. M. Cobb, P. J. S. Franks, K. Chhak, A. J. Miller, J. C. McWilliams, et al. « North Pacific Gyre Oscillation Links Ocean Climate and Ecosystem Change ». *Geophysical Research Letters* 35, n° 8 (2008). https://doi.org/10.1029/2007GL032838.

Tranchant, Benoit, Isabelle Pujol, Emanuele Di Lorenzo, et Jean-François Legeais. « The North Pacific Gyre Oscillation ». *Journal of Operational Oceanography*, Copernicus Marine Service Ocean State Report, Issue 3 (2019): s29-31. https://doi.org/10.1080/ 1755876X.2019.1633075.

Walker, G. T., et E. W. Bliss. *World Weather V*. Memoirs of the Royal Meteorological Society 4, 1932.

Yi, Daling Li, Bolan Gan, Lixin Wu, et Arthur J. Miller. « The North Pacific Gyre Oscillation and Mechanisms of Its Decadal Variability in CMIP5 Models ». *Journal of Climate* 31, n° 6 (15 mars 2018): 2487-2509. https://doi.org/10.1175/JCLI-D-17-0344.1.

4) Annex

The North Pacific Gyre Oscillation

Authors : Benoît Tranchant, Isabelle Pujol, Emanuele Di Lorenzo, Jean-François Legeais

Statement of main outcome : The North Pacific Gyre Oscillation is defined as the second dominant mode of variability of SSH anomaly and SST anomaly in the North East Pacific. This mode emerges as the leading mode of decadal variability for surface salinity and upper ocean nutrients, and is in a negative phase since 2014. This condition reflects a reduced amplitude of the central and eastern branches of the North Pacific gyre, corresponding to a reduced coastal upwelling and thus a lower sea surface salinity and concentration of nutrients. Here, this indicator is inferred from a long time-series of SSH anomalies over the North East Pacific (25°–62°N, 180°–250°E) and will be regularly updated.

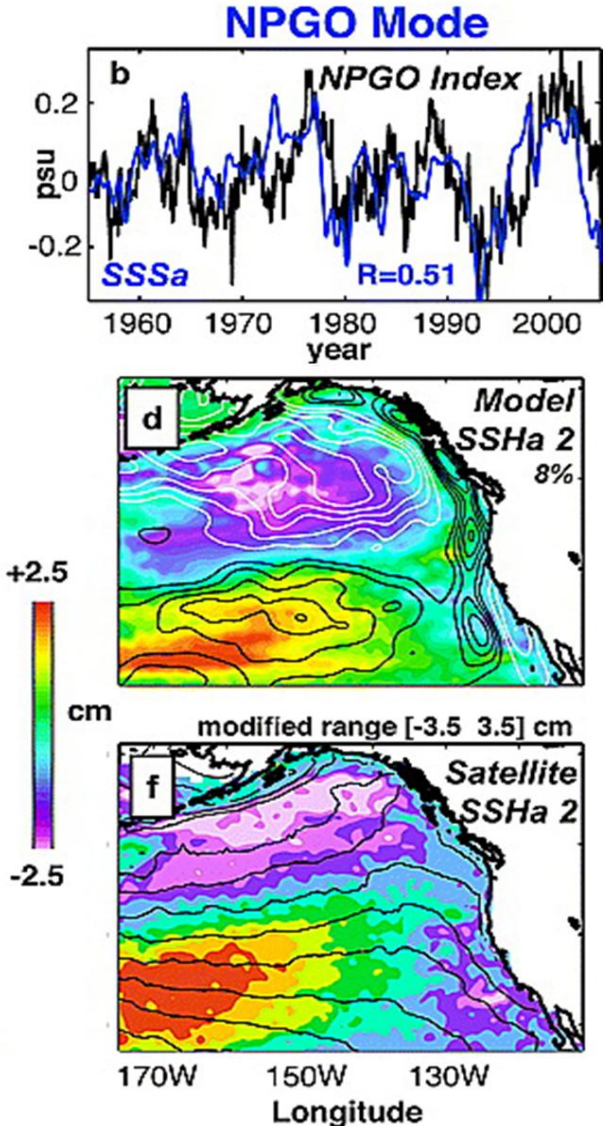
Product used :

Ref. No.	Product name & type	Documentation
2.6.1	SSH anomalies Numerical Ocean Model	(Di Lorenzo et al., 2008)
2.6.2	SSS anomalies Observations	Scripps Pier measurements collected by the Birch Aquarium at Scripps staff and volunteers. Data provided by the Shore Stations Program sponsored at Scripps Institution of Oceanography by California State Parks, Division of Boating and Waterways. Contact: shorestation@ucsd.edu.
2.6.3	NCEP wind stress 1950-2004 Numerical Atmospheric Model	The NCEP/NCAR 40-Year Reanalysis Project (Kalnay et al. 1996)

Ref. No.	Product name & type	Documentation
2.6.4	Altimeter data: previous version of SEALEVEL_GLO_PHY_L4_REP_ OBSERVATIONS_008_047 (see below) Remote sensing	Product User Manual (CMEMS-SL-PUM-008-032-051) https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-SL-PUM-008-032-051.pdf Quality Information Document (CMEMS-SL-QUID-008-032-051) https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-SL-QUID-008-032-051.pdf
2.6.5	SEALEVEL_GLO_PHY_L4_REP_ OBSERVATIONS_008_047 Remote sensing	Product User Manual (CMEMS-SL-PUM-008-032-051) https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-SL-PUM-008-032-051.pdf

		Quality Information Document (CMEMS-SL-QUID-008-032-051) https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-SL-QUID-008-032-051.pdf
2.6.6	SEALEVEL_GLO_PHY_L4_NRT_OBSERVATIONS_008_046 Remote sensing	Product User Manual (CMEMS-SL-PUM-008-032-051) https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-SL-PUM-008-032-051.pdf Quality Information Document (CMEMS-SL-QUID-008-032-051) https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-SL-QUID-008-032-051.pdf

The Decadal North Pacific Gyre Oscillation is a climate pattern defined by Di Lorenzo et al. (2008). The North Pacific Gyre Oscillation term is used because its fluctuations reflect changes in the intensity of the central and eastern branches of the North Pacific gyre circulations. Thus, it has been defined as the 2nd dominant mode of sea surface height variability in the Northeast Pacific (25°–62°N, 180°–250°E, see Figure 2.6.1d and f). The North Pacific Gyre Oscillation index has been defined as the second Principal Component of model Sea Surface Height anomaly calculated over the period 1950-2004 (reference product 2.6.1), see Figure 2.6.1 (b). Di Lorenzo et al. (2008)



suggest that the North Pacific Gyre Oscillation is the oceanic expression of the atmospheric North Pacific Oscillation (Walker and Bliss 1932). This seems reinforced by the recent work of Yi et al. (2018) that show consistent pattern feature between the atmospheric North Pacific Oscillation and the oceanic North Pacific Gyre Oscillation.

Figure 2.6.1: (b) Time series of La Jolla Scripps Pier SSSa (reference product 2.6.2) (blue) compared to the NPGO index (correlation R = 0.51, 99% significance), defined as the second PC of model SSHa (black). The SSSa records are detrended and a 12 months running mean is applied. (d) Regression maps of model NPGO indices with the model SSHa. The white (black) contours indicate regions of positive (negative) wind stress curl (Ekman upwelling) as inferred from a regression of the model PCs with the NCEP wind stress curl (reference product 2.6.3). The % variance accounted for by each regression

map is indicated. (f) Regression maps of the model NPGO index with satellite altimeter data gathered between 1993–2004 (reference product 2.6.4). The color range is modified to [3.5 3.5] in Figure 2f to properly display the map (black contours are satellite/drifter-derived mean dynamic height [Niiler et al. 2003]). Credit from Di Lorenzo et al. (2008).

This index measures change in the North Pacific gyres circulation and explains key physical-biological ocean variables including temperature, salinity, sea level, nutrients, chlorophyll-a. One example is the coherent variation of model and observed/modeled SSS (Sea Surface Salinity) in the NorthEast Pacific (Di Lorenzo et al. 2008), see Figure 2.6.1 (b). Like the Pacific decadal oscillation, the North Pacific Gyre Oscillation is a basin-scale feature and captures prominent low-frequency changes in the Pacific physical and biological fields. This is why the North Pacific Gyre Oscillation index provides a strong indicator of fluctuations in the mechanisms driving planktonic ecosystem dynamics.

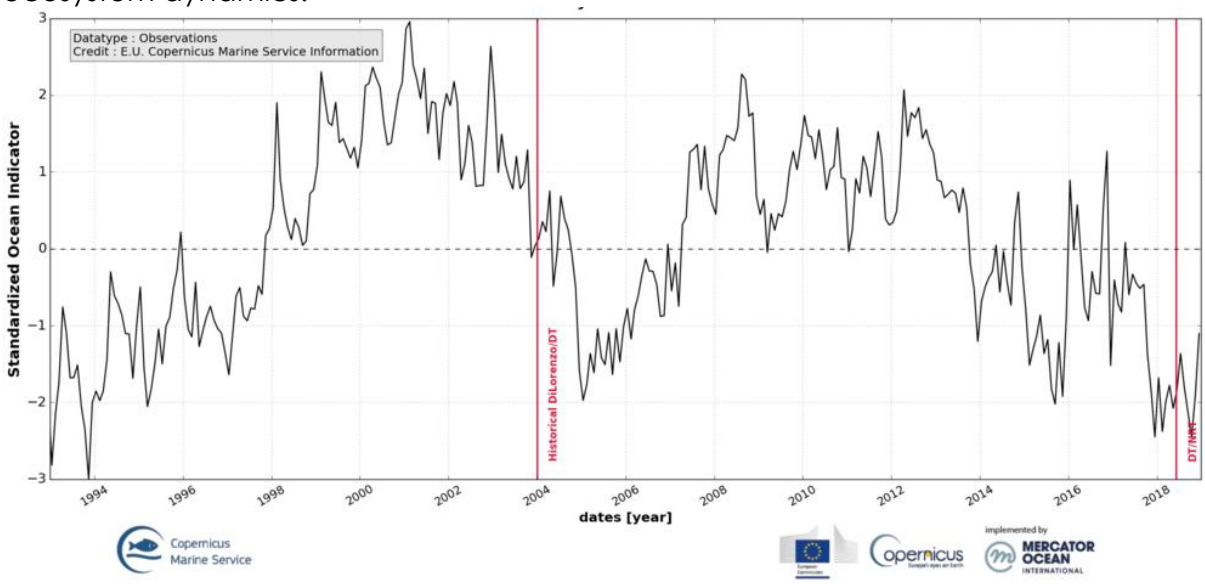


Figure 2.6.2: NPGO index monthly averages. From 1993 to 2003, the NPGO index is calculated from ref 2.6.4, see DiLorenzo et al. (2008). From 2004 onward, the NPGO index has been projected on normalized satellite altimeter Delayed-Time (DT) (CMEMS reference product 2.6.5) and Near-Real-Time (NRT) (CMEMS reference product 2.6.6).

A positive North Pacific Gyre Oscillation phase is a dipole pattern with negative SSH anomaly north of 40°N and the opposite south of 40°N. The North Pacific Gyre Oscillation index is presently in a negative phase, associated with a positive SSH anomaly north of 40°N and negative south of 40°N. This reflects a reduced amplitude of the central and eastern branches of the North Pacific gyre, corresponding to a reduced coastal upwelling and thus a lower sea surface salinity and concentration of nutrients.

The EOF's amplitude pattern (pers. Comm. Di Lorenzo) has been inferred from the principal component analysis. This regression map of the model North Pacific Gyre Oscillation index is used/projected with normalized satellite altimeter delayed-time sea level anomalies to calculate and update the NPGO index. An example is given on Figure 2.6.2 where the NPGO index has been projected on the last version of the reprocessed altimeter sea level record (CMEMS reference product 2.6.5).